Esmeralda Amado

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EDUCATION

California State University, Los Angeles (Cal State LA) ***Expected May 2027***
Bachelor of Science Computer Science GPA 3.5 | Dean's List
Minor in Biomedical Engineering

TECHNICAL SKILLS

- **Programming:** Python, C++, C, Java, JavaScript, SQL, Bash
- **Embedded & Robotics:** Arduino, ESP32, Raspberry Pi, RTOS, ROS/ROS2, VEX Robotics
- **Hardware & Prototyping:** KiCad, PCB Design, Breadboarding, FSRs, Sensors, Electrical Circuits, OnShape
- **AI & Computer Vision:** OpenCV, PyTorch, YOLO, Roboflow
- **Tools & Systems:** Linux, Docker, Git, GitHub, Wireshark, Google Colab

WORK EXPERIENCE

Software Engineering Intern & Project Leader | LA Tech Rising Technical Internship *August 2026 - Present*
- Serve as Project Leader for a software engineering team, coordinating technical planning, development tasks, milestones, and collaboration on an industry focused application.
- Contribute to software design, development, testing, and debugging while applying AI/software engineering concepts through an industry mentored project.

AI, Robotics & Engineering Instructor | iD Tech @ Caltech *June 2026 - August 2026*
- Instructed weekly cohorts of ~10 students in VEX robotics, C++, AI/ML, and engineering design through hands-on builds, programming, testing, and iterative problem solving.
- Built, tested, troubleshot, and disassembled VEX robots, gaining hands-on experience with motors, sensors, controllers, mechanical systems, and hardware-software integration.
- Led an all-girls BattleBots course and taught engineering fundamentals and Python-based AI/ML across multiple technical curricula.

Undergraduate Research Assistant | California State University, Los Angeles (Dr. Manveen Kaur's Lab) *October 2025 - Present*
- Research computer vision systems for autonomous UAV platforms used in environmental monitoring.
- Develop real-time perception pipelines using Python, OpenCV, and YOLO and curate 5,000+ labeled images using Roboflow.
- Analyze latency, sensor noise, embedded compute limitations, and real world deployment constraints in cyber-physical systems.
- Co-author research on real-world ML deployment in cyber-physical systems, focusing on edge computing, embedded systems, and simulation-to-real-world challenges.

PROJECTS

JPL-Sponsored Delay Tolerant Networking | Senior Design - Dr. Manveen Kaur *August 2026 - Present*
- Collaborate with JPL and Caltech technical mentors on a sponsored Senior Design project applying Delay Tolerant Networking (DTN) to resilient terrestrial communications.
- Implement and evaluate DTN networking algorithms for Internet2 integration across disrupted and intermittent network environments

Autonomous UAV Perception and Navigation System | ASME C-UASC Competition *August 2024 - October 2025*
Raspberry Pi 5, Roboflow, GoogleColab, Python, YoloV5, YoloV8, Yolo11
- Developed autonomous UAV perception and navigation across a 4-person computer vision team and 8-person autonomy team..
- Trained YOLO models using 5,000+ labeled images and deployed real-time object detection on a Raspberry Pi 5 using Python and OpenCV.
- Integrated perception with navigation logic and debugged latency and communication issues between perception and control subsystems.

PRD Autopilot | World Product Day 2026 Hackathon *May 2026 - June 2026*
- Collaborated on a 3-person team to develop an AI-powered platform for generating PRDs, validation reports, engineering tickets, and investor pitch decks.
- Implemented and tested user authentication workflows and conducted end-to-end QA testing across core application features.